

DevOps Engineer

Step 6: Build CI/CD Pipelines

CareerCompass — Detailed Learning Notes

What this step is about

Automate testing, building and deployment so releases are repeatable.

Core Concepts

- Continuous Integration validates changes through automated build/test/quality checks.
- Continuous Delivery/Deployment automates release steps after validation.
- Learn workflow triggers, jobs, runners, artifacts, secrets, permissions and environments.
- Use least-privilege GitHub token permissions and protected environments for production.

Skills You Should Be Able To Demonstrate

- Create a push/pull-request workflow.
- Run automated tests.
- Build a Docker image.
- Store artifacts.
- Deploy only after checks pass.

Hands-on Project / Practice

- Create a GitHub Actions pipeline that tests a web app, builds its Docker image and deploys to staging.
- Add a protected production job requiring approval.

Recommended References

- GitHub Documentation — <https://docs.github.com/>
- GitHub Actions Certification Course — <https://www.youtube.com/watch?v=Tz7FsunBbfQ>